

M52: Final Composite Dependent Variable

PCA on Raw Switch Counts (Time, Topic-LDA, Typing)

No standardization (z-score) applied

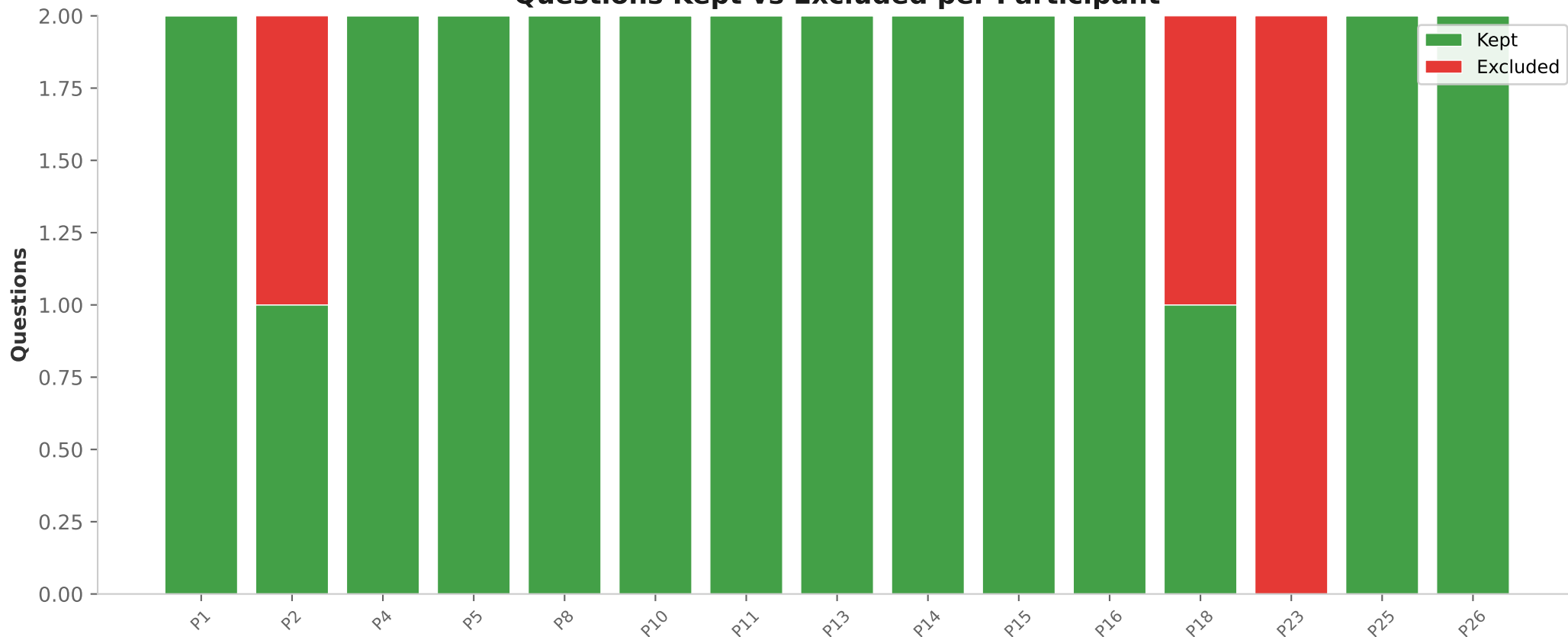
Exclusion Criteria Applied:

1. Questions with fewer than 3 page visits: 4 excluded
2. Questions with idle time $\geq 50\%$ after last meaningful event: 3 excluded
3. Participants > 3 SD from mean on any DV: 0 excluded

Result: 26 / 30 questions retained

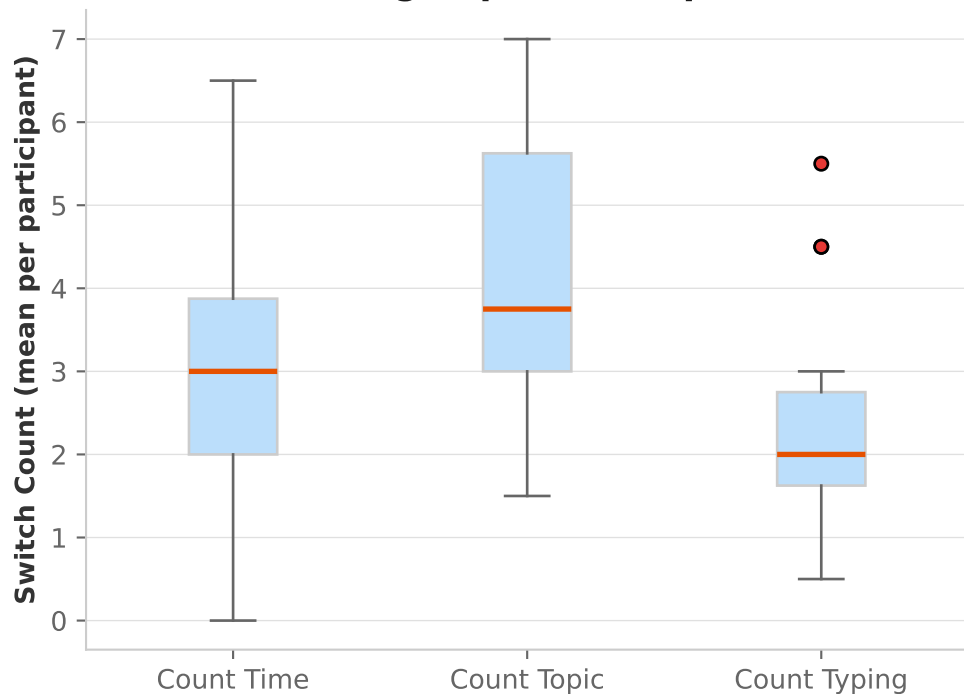
Final sample: N = 14 participants

Exclusion Summary: Questions Kept vs Excluded per Participant

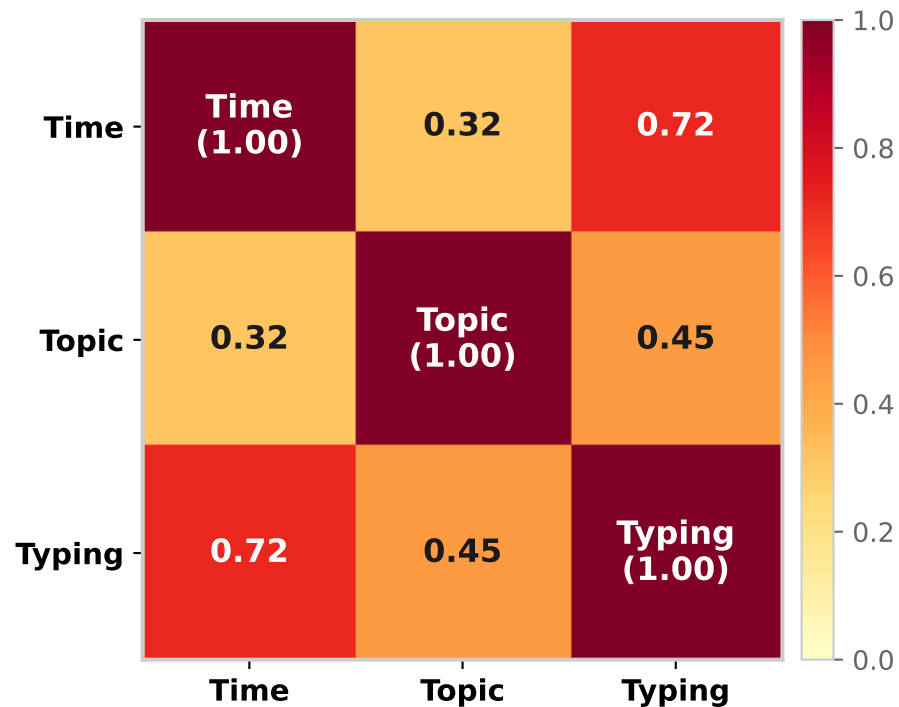


Descriptive Statistics: Raw Switch Counts

Distribution of Raw Switch Counts (Averaged per Participant)

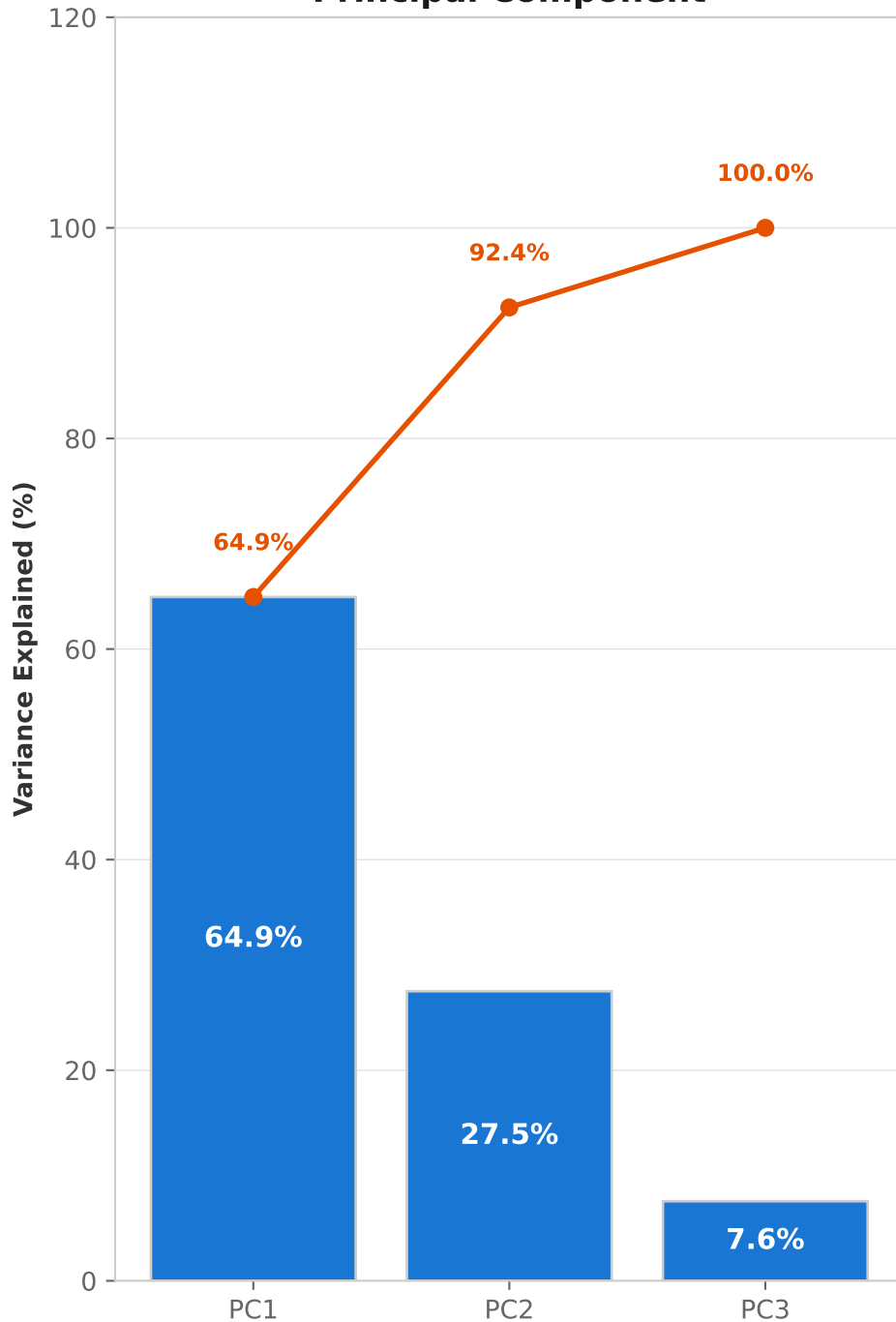


Correlation Between Switch Count Variables



PCA Results: Dimensionality Reduction on Raw Switch Counts

Variance Explained by Each Principal Component



PCA Loadings (variable weights per component):

PC1 (64.9% variance):

Count Topic	+0.640	+++++
Count Time	+0.551	+++++
Count Typing	+0.536	+++++

PC2 (27.5% variance):

Count Topic	+0.756	+++++
Count Time	-0.577	-----
Count Typing	-0.310	-----

PC3 (7.6% variance):

Count Typing	+0.786	+++++
Count Time	-0.603	-----
Count Topic	-0.138	--

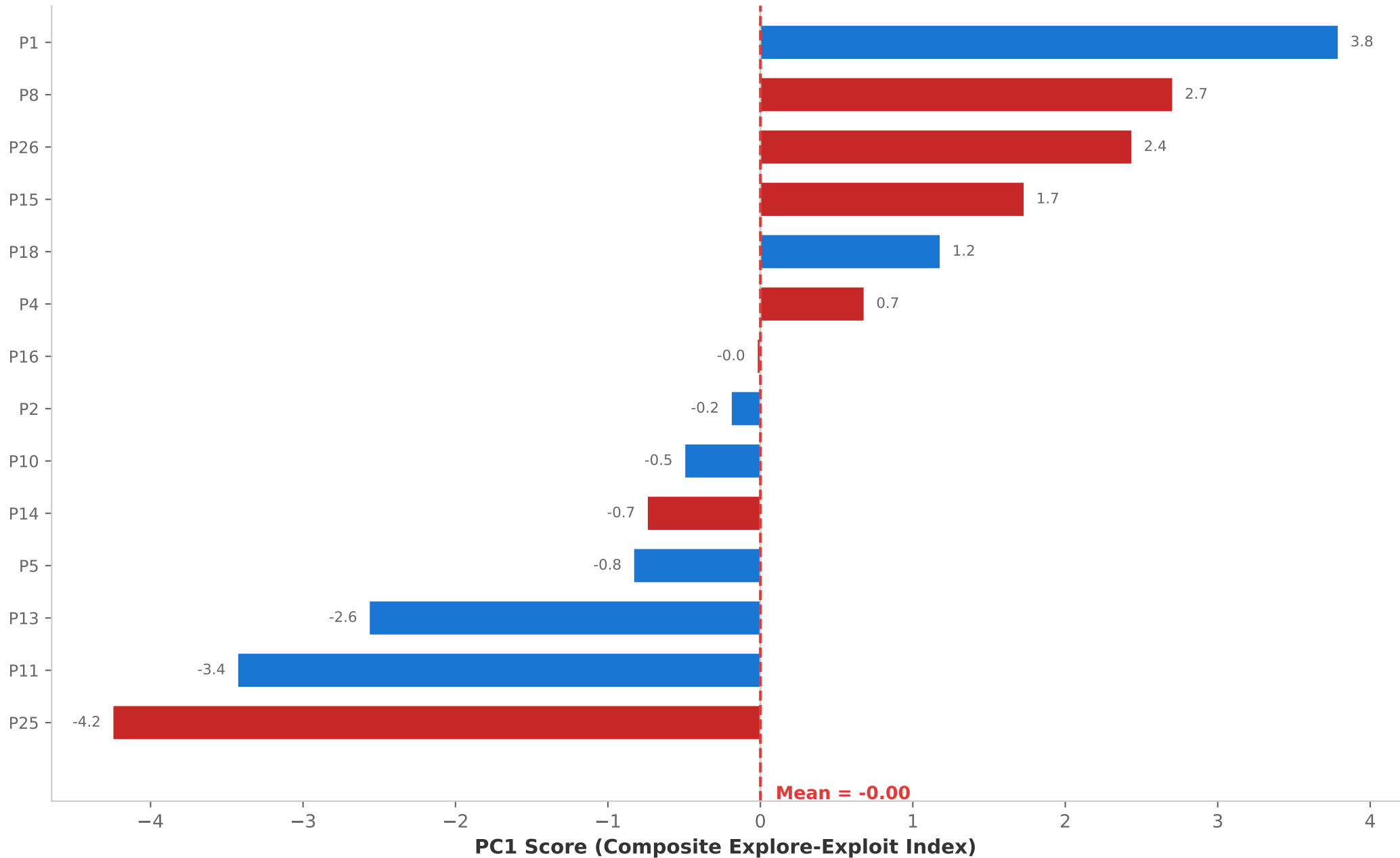
Interpretation:

PC1: Primarily reflects topic switching
(loading = +0.640)
Higher score = more topic transitions

PC2: Primarily reflects topic switching
(loading = +0.756)
Higher score = more topic transitions

PC3: Primarily reflects typing switching
(loading = +0.786)
Higher score = more typing transitions

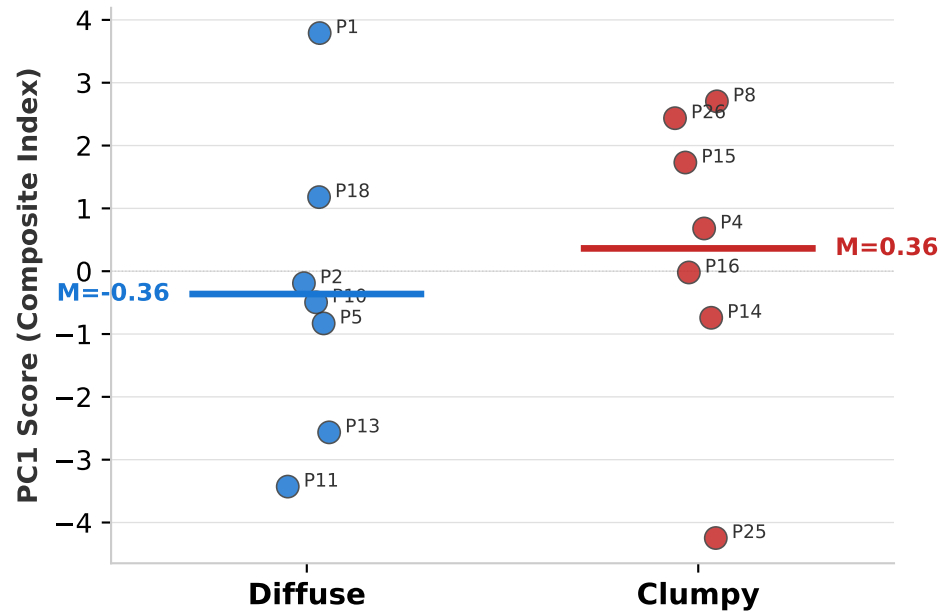
Individual PC1 Scores: Composite Explore-Exploit Index



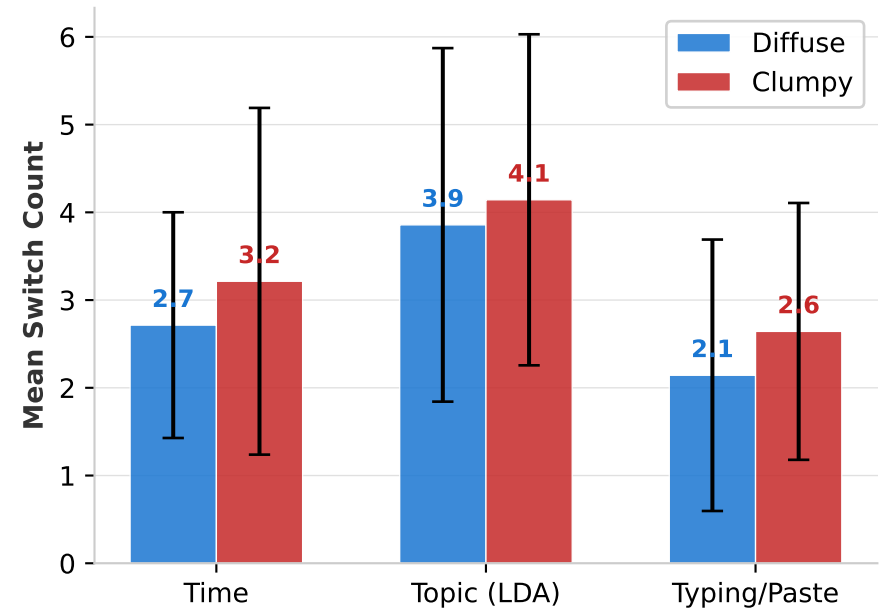
Condition Comparison: Clumpy vs Diffuse

N = 14 (Diffuse: 7, Clumpy: 7)

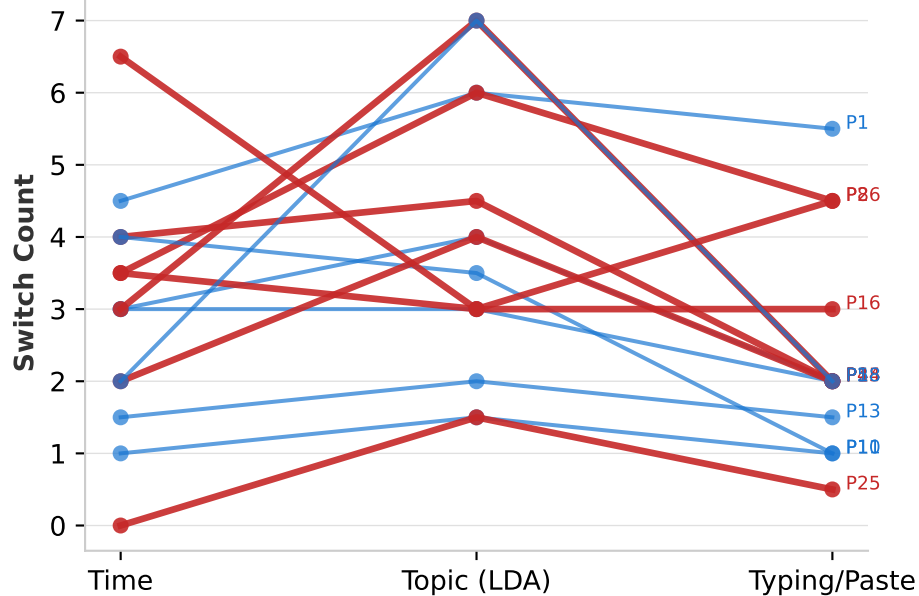
PC1: Composite Explore-Exploit
by Condition



Raw Switch Counts
by Condition (Mean +/- SD)



Individual Profiles
by Switch Type



Condition Comparison: Descriptive Statistics & t-Tests

N = 14 (Diffuse: 7, Clumpy: 7)

DESCRIPTIVE STATISTICS

Variable	Diffuse (n=7)	Clumpy (n=7)
Time	M = 2.71, SD = 1.29	M = 3.21, SD = 1.98
Topic (LDA)	M = 3.86, SD = 2.01	M = 4.14, SD = 1.89
Typing/Paste	M = 2.14, SD = 1.55	M = 2.64, SD = 1.46
PC1 (composite)	M = -0.36, SD = 2.39	M = 0.36, SD = 2.39

INDEPENDENT-SAMPLES t-TESTS (Welch)

Variable	t	df	p	Cohen d	Sig.
Time	-0.56	10.3	0.5867	+0.30	n.s.
Topic (LDA)	-0.27	11.9	0.7889	+0.15	n.s.
Typing/Paste	-0.62	12.0	0.5462	+0.33	n.s.
PC1 (composite)	-0.57	12.0	0.5802	+0.30	n.s.

* p < .05 ** p < .01 *** p < .001

INTERPRETATION

PC1 difference (Clumpy - Diffuse): +0.73
Clumpy condition shows more explore-exploit switches
Effect size (Cohen's d): +0.30
Result is NOT statistically significant (p = 0.5802)
Low statistical power due to small sample (N = 14)

Largest sub-variable effect: Typing/Paste
d = +0.33, p = 0.5462

Note: Typing metric relies on paste events only
(answer_snapshot missing from current log).